

PRODUCT REQUIREMENTS DOCUMENT

Ammlan Rath — Portfolio Website

An implementation-ready, enterprise-grade specification engineered to remove ambiguity for AI coding assistants and human development teams. Defines exactly what must be built and how every component, page, and interaction must behave.

DOCUMENT TYPE	Enterprise Product Requirements Document (PRD) — Implementation Edition
VERSION	2.0 — Final, Implementation-Ready
STATUS	Approved for Development — Zero-Ambiguity Baseline
PREPARED BY	Product & UX Architecture Office
PREPARED FOR	Ammlan Rath — Owner / Stakeholder; AI Coding Assistants; Development Team
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SUPERSEDES	PRD v1.0 (June 2026) — fully incorporated and expanded herein
CLASSIFICATION	Internal — Development Reference Copy

Revision History & Document Control

Revision History

Version	Date	Author	Description of Change
0.1	May 2026	Product & UX Architecture Office	Initial draft — project overview, tech stack, folder structure captured from stakeholder brief.
0.5	June 2026	Product & UX Architecture Office	Added information architecture, page-level specifications, and design system.
0.9	June 2026	Product & UX Architecture Office	Added animation specification chapter, SEO chapter, accessibility and performance chapters.
1.0	July 2026	Product & UX Architecture Office	Final formatting pass, appendices, glossary, and roadmap added. Approved for development.
2.0	July 2026	Product & UX Architecture Office	Implementation-Ready Edition. Added: complete component inventory (35+ components) with full state/interaction/ accessibility specs; per-page structural wireframes across four breakpoints; complete user-flow documentation for six personas; typography, spacing, design-token, and motion systems; image specification catalogue; content & voice guidelines; expanded per-page SEO; measurable performance budget; expanded accessibility implementation guide; measurable acceptance criteria per page/feature; functional behaviour specification for every interactive element; page-by-page functional breakdown chapter; expanded future-scalability roadmap. Document restructured for zero-ambiguity AI-assisted implementation.

Document Purpose

This Product Requirements Document (PRD) is the single source of truth for the design and development of the Ammlan Rath Portfolio Website. Version 2.0 exists specifically to close every remaining interpretive gap in v1.0 so that a UI/UX designer, a frontend developer, a backend engineer (for future phases), or an AI coding assistant can implement the entire product end-to-end **without needing to ask the stakeholder any clarifying questions**. Every page, component, state, interaction, animation, content rule, and acceptance criterion referenced anywhere in this document is a binding requirement unless explicitly marked as a recommendation (**RECOMMENDATION**) or future consideration. Binding requirements are marked **REQUIRED** where emphasis is useful.

Priority Order (governs every trade-off decision in this document): 1. Productivity 2. Usability 3. Functional Effectiveness 4. Technical Correctness 5. Scalability 6. Maintainability 7. Design Consistency 8. Visual Design. Visual/aesthetic decisions are intentionally kept to "modern and clean" — this document favors precise functional, structural, and behavioral specification over decorative description.

How to Use This Document

Reader	Recommended Reading Path
UI/UX Designer	Part I (Context), Part II (Design System), Part III (Component Library), Part IV (Page Specifications & Wireframes)
Frontend Developer	Part II (Design System), Part III (Component Library incl. states/interactions), Part IV (Page Specifications), Part V (Functional Behaviour Spec), Part VI (Animation), Part VII (Responsiveness), Part VIII (Accessibility)
AI Coding Assistant	The entire document, in order. Every chapter is self-contained, cross-referenced, and written to remove implicit assumptions.
Stakeholder / Owner	Part I (Sections 1–8), Part IX (Acceptance Criteria), Part XII (Roadmap)
QA / Reviewer	Part IX (Acceptance Criteria), Part V (Functional Behaviour Spec), Part VIII (Accessibility)

NOTE ON SCOPE: This PRD defines requirements only. It intentionally contains no source code, no markup, and no scripting. All implementation decisions must derive from the specifications herein, not from external inspiration or assumptions. Where this document is silent, the implementer must treat the gap as an Open Question (Appendix C) rather than infer a decision.

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Part I — Product Context (Consolidated from v1.0)

This part consolidates the strategic context established in PRD v1.0 (Executive Summary through Directory Structure) in condensed form. It remains fully binding. Full narrative detail is preserved; where v1.0 content is unchanged it is summarized here for continuity, and the reader is pointed to the expanded chapters (Parts II–XII) that supersede v1.0's shallower treatment of design, components, and page behaviour.

1. Executive Summary

Ammlan Rath's portfolio website is a premium, multi-page, animation-rich showcase built entirely with HTML5, CSS3, and vanilla JavaScript — no frameworks, no CSS libraries, no backend. It presents education, skills, projects, blog writing, and freelance services with the polish of an award-winning creative-agency site rather than a conventional developer portfolio. Every visitor — recruiter, client, or peer developer — must be able to form a clear impression within ten seconds and reach any deep content within three clicks.

The site comprises nine primary destinations (Home, About, Education, Projects, Project Detail, Blog, Blog Detail, Hire Me, Contact) plus seven error/status pages. All content is static, template-driven, and built for unlimited future growth. Version 2.0 of this document adds the complete component inventory, wireframes, user flows, interaction/state specifications, design-token and motion systems, image specifications, content guidelines, per-page SEO, performance budgets, accessibility implementation detail, measurable acceptance criteria, and a full functional-behaviour specification — closing every ambiguity left open in v1.0.

2. Project Vision

The portfolio functions as Ammlan Rath's primary professional identity on the internet, built to read like a cinematic digital-agency site rather than a link list. It is architected for unlimited content growth using static templates and a strictly separated content/presentation model, so it can later absorb a CMS or headless backend without visual or navigational change.

3. Problem Statement

Problem	Typical Symptom	How This Product Solves It
Visual sameness	Template-driven layouts, default fonts/spacing, interchangeable hero sections.	Bespoke luxury dark design system — Part II.
Flat, static presentation	No motion, no storytelling, uniform hierarchy.	Dedicated motion system — Part VI.
Weak storytelling	Resume-style About pages; screenshot-only project pages.	Narrative About page; Problem/Solution structured Project Detail — Part IV.
Poor scalability	Redesign required to add content.	Strict template-driven content model — Section 16, Part XII.

4. Project Goals

#	Goal	Success Signal
PG-01	Deliver a premium, agency-grade visual identity	Visual review confirms no resemblance to a default framework theme
PG-02	Present all core professional content	Education, projects, blogs, services, contact all reachable and complete
PG-03	Deliver a smooth, animation-rich experience	Every transition/scroll/hover follows Part VI timing rules
PG-04	Remain fully responsive	All pages correct across four breakpoints, both orientations — Part VII
PG-05	Be built for unlimited content growth	New entries addable via existing templates, zero structural change
PG-06	Achieve strong SEO fundamentals	Every page has correct meta/OG/schema data — Part X
PG-07	Meet WCAG 2.1 AA	Keyboard, focus, contrast requirements met — Part VIII
PG-08	Load fast despite heavy visuals	Performance budget in Part XI met on mid-range mobile

5. Business Goals

#	Business Goal	Rationale
BG-01	Increase inbound recruiter interest	Premium, memorable site differentiates from generic candidates
BG-02	Generate qualified freelance leads	Hire Me page and inquiry flow convert visitors to conversations
BG-03	Establish long-term personal brand equity	Consistent identity + running blog build authority over time
BG-04	Reduce friction in hiring/vetting	Detailed case studies let evaluators assess without a live interview
BG-05	Support long-term SEO discoverability	Structured metadata + blog strategy compound organic visibility

6. Target Audience & 7. User Personas (Summary)

Persona	Primary Goal	Key Pages
Priya — Technical Recruiter	Screen 15–20 portfolios/week in under 2 minutes each	Home, About, Education, Projects
David — Startup Founder / Freelance Client	Judge design taste, then pricing/ process	Home, Hire Me, Contact
Meera — CS Student	Find inspiration and practical guidance	Blog, Education, About
	Judge real technical depth	Projects, Project Detail

Arjun — Senior Engineer / Peer		
Company / Hiring Team	Evaluate technical depth and communication	Projects, Project Detail, Education
General Visitor	Browse casually, be impressed	Home

Full complete user-flow documentation (entry point, goals, navigation path, decision points, primary/secondary actions, exit points) for every persona listed above is provided in **Part V — Complete User Flow Documentation**.

8. User Stories & 9. Functional Requirements & 10. Non-Functional Requirements

Retained in full from v1.0 (US-01 through US-12, FR-01 through FR-14, and the Non-Functional Requirements table covering Performance, Responsiveness, Accessibility, Maintainability, Portability, Scalability, Browser Support, and Visual Consistency). These are unchanged and remain binding. Version 2.0 makes each of these requirements independently testable via **Part IX — Acceptance Criteria** and **Part V — Functional Behaviour Specification**, which decompose every requirement into per-element, per-page pass/fail conditions.

11–15. Information Architecture, Navigation, Sitemap, Page Hierarchy, Directory Structure

Retained in full from v1.0 Sections 11–16 (Content Groupings, global navigation model, Navigation Flow Diagram, complete route tree, page hierarchy levels, and the required project directory layout with rationale table). These are unchanged and remain binding infrastructure decisions. No page, component, or asset introduced anywhere in Parts II–XII of this document may violate the directory structure or route tree defined in v1.0 Section 16.

Reference — Directory root remains: `index.html`, `/html/` (about, education, hire, contact, project, blog, projects/, blogs/, errors/), `/assets/` (css/, js/, images/, resume/), `favicon.ico`, `robots.txt`, `sitemap.xml`. Nine primary routes, unlimited project/blog child routes, seven error routes.

Part II — Design System (Expanded)

This part expands v1.0 Section 16 into four independently implementable systems: Design Tokens, Typography, Spacing/Grid, and Motion. No page or component may define a value outside these systems.

16. Design Tokens

Tokens are the atomic, named values every stylesheet must reference — never a hard-coded hex, px, or ms value outside `global.css :root`. Token names below are binding; exact hex/rgba values are implementation-selectable within the character described, subject to the contrast rules in Part VIII.

Token	Category	Definition / Usage Rule
<code>--color-primary</code>	Brand	The single accent gold/bronze hue. Used for links, active states, key CTAs, section markers, focus rings. Exactly one accent hue site-wide (v1.0 constraint) — required.
<code>--color-secondary</code>	Brand	A muted neutral (grey-blue) used for secondary UI chrome — dividers, disabled states, inactive icons.
<code>--color-accent</code>	Brand	Alias of <code>--color-primary</code> ; retained as a separate token name only for semantic clarity in component code (e.g. badge accents). Must resolve to the same value as <code>--color-primary</code> .
<code>--color-bg</code>	Surface	Primary page background — deep charcoal-black, not pure #000.
<code>--color-surface</code>	Surface	Elevated surface (cards, panels, modals) — one step lighter than <code>--color-bg</code> .
<code>--color-card</code>	Surface	Card-specific surface value; may equal <code>--color-surface</code> or be a distinct, slightly warmer shade to differentiate cards from modals.
<code>--color-border</code>	Surface	Hairline dividers/card outlines — low-opacity off-white, barely visible except hover/focus.
<code>--color-text-primary</code>	Text	Headlines/body — off-white, never pure #FFFFFF.
<code>--color-text-secondary</code>	Text	Supporting copy/captions/meta — 60–70% brightness of primary text.
<code>--color-hover</code>	Interactive	Hover-state overlay/shift, typically a low-opacity tint of <code>--color-primary</code> applied on top of the resting surface/text color.
<code>--color-focus</code>	Interactive	Focus-ring color; must equal <code>--color-primary</code> for visual consistency and must meet 3:1 contrast against adjacent surfaces (Part VIII).
<code>--color-success</code>	Status	Form validation success only — desaturated green.
<code>--color-warning</code>	Status	Non-blocking warnings (e.g. "field will be validated") — desaturated amber.
<code>--color-danger</code>	Status	Form validation error / destructive states — desaturated red.
<code>--color-info</code>	Status	Neutral informational banners (e.g. Maintenance page) — desaturated blue-grey, never the accent gold.

All status colors must remain desaturated enough not to clash with the luxury dark palette, and must never be the sole indicator of state — always paired with icon and/or text per Part VIII.

17. Typography System

Style	Desktop Size	Weight	Line Height	Where Used
H1 — Page Title	40–56px	700	1.15	Exactly one per page; the page's primary heading, always the true semantic <h1>.
Display / Hero	64–96px	700–800	1.05	Home hero headline and section-opening statements only; never used more than once per page.
H2 — Section Heading	28–34px	600–700	1.2	Major page sections (e.g. "Featured Projects", "Services").
H3 — Subheading	18–22px	600	1.3	Card titles, sub-sections within an H2 block.
H4 / H5 / H6	15–17px / 14px / 13px	600	1.35	Nested content structure inside long-form pages (About, Blog Detail) where a third or fourth heading level is required for strict semantic order; never skipped (Part VIII).
Subtitle	18–20px	500	1.4	Directly beneath an H1/H2 as a one-line elaboration (e.g. hero role statement).
Body Large	18px	400	1.7	Lead paragraphs — About narrative opening, Blog Detail opening paragraph.
Body Regular	16px	400	1.6	Default paragraph text across all pages.
Body Small	14px	400	1.55	Secondary descriptive text (card descriptions, FAQ answers).
Caption / Meta	12–13px	500	1.4	Dates, tags, reading time, image captions, form helper text.
Label	12px	600	1.2	Form field labels, filter chip labels — uppercase, letter-spacing 0.5–1px.
Button Text	14–16px	600	1	All button/CTA text, no exceptions.
Navigation Text	15px	500	1	Header/footer navigation links.
Code Text	14px	400	1.5	Monospaced font family; used only inside Blog Detail code blocks, never for site UI.
Quote	22–28px	500, italic	1.4	Pull-quote blocks on About and select Blog Detail moments.

Two typefaces maximum: one editorial sans-serif for H1/H2/Display/Quote, one highly legible sans-serif for everything else. Maximum two weights per family in active use (plus 700/800 for Display). Mobile scale reduces

every size above by approximately 35–45% per Part VII breakpoint rules, never below 16px for Body Regular (a hard accessibility floor — Part VIII).

18. Spacing & Grid System

Token	Value	Usage Rule
--space-unit	8px	Base unit. Every spacing value site-wide must be a multiple of 8 (4px permitted only for icon-to-label micro-gaps).
Margin Scale	8 / 16 / 24 / 32 / 48 / 64 / 96px	Named scale steps: xs, sm, md, lg, xl, 2xl, 3xl. Component margins must snap to a scale step, never an arbitrary value.
Padding Scale	Same 7-step scale as Margin	Card internal padding defaults to md–lg (24–32px) desktop, sm–md (16–24px) mobile.
Section Vertical Rhythm	96–160px desktop, 56–88px mobile	Space between major page sections (Section 18.1.5 pattern), reduced proportionally at each breakpoint (Part VII).
Card Spacing (grid gutter)	24–32px desktop, 16–20px mobile	Space between grid cards; must equal the Grid Gutter token below for visual consistency.
Container Max-Width	1200–1320px	All page content is constrained to this width and horizontally centered; full-bleed exceptions permitted only for Hero backgrounds and the Freelance CTA band.
Grid — Desktop/Laptop	12 columns	Card grids default to 3–4 columns (Projects/Blog) or 4–6 columns (Skills) depending on card width.
Grid — Tablet	6 columns	Card grids reduce to 2 columns.
Grid — Mobile	1 column	All grids collapse to a single column; horizontal scrolling grids are prohibited except the optional Skills marquee.
Gutter	24–32px desktop, 16px mobile	Fixed gutter width applied uniformly across every grid on the site.
Alignment Rule	—	All text blocks left-align by default; only Hero, Freelance CTA band, and error-page messaging may center-align. Right-alignment is prohibited site-wide.

19. Motion System (Governance Layer)

The complete per-animation table (trigger, duration, easing, purpose, experience) is retained unchanged from v1.0 Section 18/19 and is the binding animation inventory. This section adds the governance rules that make the system implementable without guesswork:

Rule	Specification
Animation Duration Ceiling	No animation exceeds 1.5s outside the page loader and hero entrance sequence.
Animation Curve Vocabulary	Exactly three easing curves are permitted site-wide: <code>ease-out</code> (entrances/reveals), <code>ease-in-out</code> (hovers/transitions/filters), and <code>linear</code> (scroll-tied parallax/timeline-draw only). No custom cubic-bezier overshoot/bounce curves.
Animation Triggers	

	Exactly four trigger classes: (1) page load, (2) scroll-into-view (fires once per page view), (3) pointer/keyboard interaction (hover, focus, press, click), (4) continuous scroll position (parallax, timeline draw only).
Performance Considerations	Only <code>transform</code> and <code>opacity</code> may be animated for any motion lasting under 1s; properties that trigger layout reflow (width, height, top, left, margin) must never be animated.
Reduced Motion	Every animation is suppressed or reduced to an instant/near-instant ($\leq 80\text{ms}$ opacity) state change when <code>prefers-reduced-motion: reduce</code> is set. Content must be fully present and usable with motion off.
Re-trigger Rule	Scroll-triggered reveals fire once per page view; re-scrolling past an already-revealed section never re-triggers it.

Part III — Complete Component Inventory

Every reusable UI component on the site is defined below with Purpose, Responsibilities, Inputs, Outputs, Variants, Responsive Behaviour, Accessibility, Dependencies, Reusability, full State inventory, and full Interaction specification (hover/focus/active/loading/disabled/success/failure/empty/keyboard/touch/mouse). Components sharing a structural pattern are grouped into one family table with shared rows and a differentiation row; this is intentional for consistency, not a simplification of requirements — every named component still has independently defined content fields and dependencies.

20.1 Navigation Components

Navbar (Primary Header Navigation)

Purpose	Provide constant, site-wide wayfinding and brand anchoring on every page.
Responsibilities	Render logo/wordmark, primary links (Home, About, Education, Projects, Blog, Hire Me), distinct Contact CTA; manage sticky/scrolled state; manage mobile hamburger toggle.
Inputs	Current route (for active-link highlighting), scroll position, viewport width.
Outputs	Navigation events (route change), mobile-menu open/close state.
Variants	Default (top of page), Scrolled/Sticky (compact, elevated shadow), Error-Page Simplified (logo only), Mobile Collapsed.
States	Default; Scrolled (sticky, reduced height, background opacity increases); Active-link (accent underline/color on current page); Mobile-Open (full-screen overlay); Mobile-Closed; Hover (per-link underline grow); Focus (visible ring).
Interactions	Hover: link underline animates in, 150–200ms. Focus: visible accent ring, same as hover for parity. Active/Press: slight opacity dip. Keyboard: full Tab order left-to-right, Enter activates, Escape closes mobile overlay. Touch: tap opens/closes hamburger overlay with 300–450ms slide/fade. Mouse: hover-only underline; no drag behavior.
Responsive Behaviour	Desktop/Laptop: full horizontal list. Tablet: condenses to icon+label or collapses depending on item count. Mobile: hamburger + full-screen overlay (Section 20 breakpoint table, v1.0).
Accessibility	<code><nav aria-label="Primary"></code> ; current page marked <code>aria-current="page"</code> ; hamburger button has <code>aria-expanded</code> and <code>aria-controls</code> ; overlay traps focus while open and returns focus to the hamburger on close.
Dependencies	Primary Button (Contact CTA), Logo asset, global.js scroll listener.
Reusability	Single instance, included identically on all 9 primary pages and (simplified variant) on all 7 error pages.

Footer

Purpose	Provide a secondary, always-available exit to every major destination plus social/contact channels on every page.
Responsibilities	Render secondary nav list, Social Links component, copyright line, closing tagline.
Inputs	Current year (for copyright), social profile URLs.
Outputs	Navigation/outbound-link events.

Variants	Full (all primary pages), Minimal (error pages — nav only, no tagline band).
States	Default; Link Hover; Link Focus.
Interactions	Hover/Focus: underline or color shift on links, 150–200ms, identical to Navbar link treatment for consistency. Keyboard: standard Tab order. Touch/Mouse: standard tap/click, no special gestures.
Responsive Behaviour	Desktop: multi-column link groups + social row. Tablet: 2-column. Mobile: single stacked column, social icons centered.
Accessibility	<code><footer></code> landmark; link list uses semantic <code><nav aria-label="Footer"></code> .
Dependencies	Social Links component.
Reusability	Single instance, present identically (content-wise) on all 16 routes.

Breadcrumb-Equivalent ("Back to X")

Purpose	Give Project Detail and Blog Detail pages a lightweight sense of place within their parent listing.
Responsibilities	Render a single link with a back-arrow icon and label ("Back to Projects" / "Back to Blog").
Inputs	Parent page route.
Outputs	Navigation event to parent listing.
Variants	Projects variant, Blog variant (label text differs only).
States	Default; Hover (arrow nudges left 2–4px); Focus (visible ring).
Interactions	Hover: icon micro-shift, 150ms ease-out. Keyboard: reachable before the hero title in Tab order. Touch: minimum 44x44px tap target.
Responsive Behaviour	Unchanged position/size across breakpoints; label may truncate to icon-only under 360px width.
Accessibility	Descriptive link text (not "click here"); icon marked <code>aria-hidden="true"</code> since the label carries the meaning.
Dependencies	Icon System.
Reusability	Used on Project Detail and Blog Detail only.

Back To Top Button

Purpose	Let visitors return to the top of long-scroll pages without repeated manual scrolling.
Responsibilities	Appear after a scroll threshold; scroll the viewport smoothly to top on activation.
Inputs	Scroll position (threshold: appears after 1 full viewport height of scroll).
Outputs	Scroll-to-top action.
Variants	Single variant, fixed bottom-right.
States	Hidden (below threshold); Visible (fade+scale in); Hover (lift + glow); Focus; Active/Press.
Interactions	

	Appear/Disappear: 200–300ms opacity+scale fade tied to scroll threshold, not continuous parallax. Hover: Card Hover timing (250–350ms). Keyboard: focusable only when visible; Enter/Space triggers smooth scroll. Touch: standard tap.
Responsive Behaviour	Present at all breakpoints; repositioned to avoid overlapping mobile nav hamburger.
Accessibility	<code>aria-label="Back to top"</code> ; respects reduced-motion (jump instead of smooth-scroll).
Dependencies	Icon System.
Reusability	Global, all long-scroll pages (Home, About, Education, Projects, Blog listing pages, Project/ Blog Detail).

20.2 Button Family — Primary / Secondary / Outline / CTA

Field	Primary Button	Secondary Button	Outline Button	CTA (Band-level)
Purpose	Highest-priority action on a view	Lower-priority alternative action	Tertiary action needing lower visual weight	Full-width conversion prompt (Hero, Freelance CTA band)
Responsibilities	Render label + optional icon; route to internal page, trigger in-page scroll, or submit a form; expose loading/disabled state when the triggered action is asynchronous.			
Inputs	Label text (required verb-first string, e.g. "View Project"), destination (route/anchor/submit handler), optional icon, disabled flag.			
Outputs	Click/activation event carrying destination or submit intent.			
Variants	Solid accent fill	Text-only / ghost outline	Accent outline, transparent fill	Large, full-width, high-contrast background band
States	Default, Hover, Focus, Active/Press, Loading (spinner replaces label, button width locked to prevent layout shift), Disabled (reduced opacity, no pointer events, <code>aria-disabled</code>), Success (brief checkmark swap for confirm-style buttons only, e.g. form submit).			
Interactions	Hover: scale 1.0 → 1.02–1.04 + glow/color shift, 150–200ms ease-out. Focus: visible accent ring, identical timing to hover. Active/Press: scale compresses slightly below resting state. Keyboard: Enter/Space activates. Touch: minimum 44x44px target, no hover-dependent affordance required to operate. Failure: button returns to Default with an adjacent inline error message (never the button itself turning red).			
Responsive	Desktop: inline, auto-width. Mobile: CTA groups stack vertically and may expand to full container width.			
Accessibility	Real <code><button></code> or <code><a></code> element as semantically appropriate (button for actions, anchor for navigation); label never icon-only without <code>aria-label</code> .			
Dependencies	Icon System, Design Tokens (color/spacing).			
Reusability	Used on every page; CTA band variant used only on Home and Hire Me.			

20.3 Card Family — Project / Blog / Certificate / Achievement / Skill / Profile / Image

Field	Specification (applies to all Card variants unless noted)
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Purpose	Present one discrete content unit (a project, post, credential, achievement, skill, or profile) as a scannable, clickable summary that routes to fuller detail or performs a lightweight in-place action.
Responsibilities	Render media/icon, title, meta row (tags/date/category as applicable), short description; expose the full card as a single click/tap target routing to detail (Project Card, Blog Card) or performing an inline action (Skill Card tooltip, Profile Card link-out).
Inputs	Content object specific to the card type — see per-type table below.
Outputs	Navigation event (Project/Blog/Certificate/Achievement cards) or no-op hover affordance (Skill card).
Variants	Grid card (default), Featured card (larger, used once per listing — Blog listing's Featured Blog), Compact card (Home preview density).
States	Default; Hover (lift + shadow + accent border glow, Card Hover timing 250–350ms); Focus (identical visual treatment to Hover, keyboard-triggered); Selected (Skill Card only — persistent accent border after tap on touch devices); Loading (skeleton card, see Skeleton Loader); Disabled (Certificate/Achievement cards with no linked asset — reduced opacity, no hover affordance, no click target); Error (thumbnail failed to load — falls back to a branded placeholder graphic, never a broken-image icon); Empty (grid-level, not card-level — see Empty State below).
Interactions	Hover: lift 4–8px, shadow increase, accent border glow, 250–350ms ease-in-out. Focus: same visual as hover, ring added. Active/Press: lift slightly compresses. Keyboard: whole card is one focusable unit (not multiple nested tab-stops); Enter activates. Touch: tap navigates directly (no separate hover step required); overlay "View" affordance shown permanently on touch devices rather than only on hover. Mouse: hover overlay fades in, 250ms.
Responsive Behaviour	Desktop/Laptop: 3–4 column grid. Tablet: 2 column. Mobile: 1 column, full container width, image aspect ratio locked to prevent CLS.
Accessibility	Entire card wrapped in a single <code><a></code> where the primary action is navigation; heading level inside the card is one level below the section's H2; image alt text is descriptive, not "thumbnail".
Dependencies	Tag/Badge component, Icon System, Image Specifications (Part... Images).
Reusability	Project Card: Home, Projects listing, Project Detail (Related Projects). Blog Card: Home, Blog listing, Blog Detail (Related Blogs). Certificate/Achievement Card: Education only. Skill Card: Home, Education. Profile Card: not used site-wide by default — reserved for future team/collaborator content (Part XII). Image Card: generic wrapper reused wherever a standalone image needs card chrome (e.g. Gallery thumbnails).

Card Type	Unique Content Fields	Click Destination
Project Card	Thumbnail, title, one-line description, 2–4 tech tags, category label	Project Detail page
Blog Card	Cover image, title, category tag, reading time, publish date, excerpt, 1–3 tags	Blog Detail page
Certificate Card	Title, issuing organization, date issued, credential image/link	Lightbox (credential image) or external verification link, new tab
Achievement Card	Title, one-line context sentence, date	None — informational only, no click target
Skill Card	Icon, skill label, optional proficiency indicator	None — hover/focus tooltip only
Profile Card	Photo, name, role/title, short bio, social links	Reserved for future multi-person content

Image Card	Image, optional caption	Lightbox (enlarge) where used in Gallery
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20.4 Content & Layout Primitives

Hero

Purpose	Establish page identity and primary action within the first viewport of Home, and provide a consistent title/context band on all other primary pages.
Responsibilities	Render headline (Display or H1 scale), optional subtext, optional primary/secondary CTAs, optional background treatment (gradient motion, parallax image).
Inputs	Headline text, subtext, CTA config, background asset (optional).
Outputs	CTA click events.
Variants	Full-viewport Home Hero (with background motion + 2 CTAs), Compact Page Hero (About/Education/Projects/Blog/Hire Me/Contact — headline + one-line subtext only), Detail Hero (Project/Blog Detail — headline + meta row + media).
Responsive Behaviour	Full-viewport variant reduces height on tablet/mobile landscape to avoid excessive scroll-before-content (v1.0 Section 20 rule).
Accessibility	Headline is always a true, unique <code><h1></code> ; background motion respects reduced-motion.
Dependencies	Primary/Secondary Button, Icon System (scroll-hint).
Reusability	One instance per page, content and variant differ per Part IV page specification.

Section Header

Purpose	Introduce every major page section with a consistent title + optional subtext pattern, reinforcing the site's visual rhythm.
Responsibilities	Render H2 title, optional one-line subtext, optional "View All" link aligned to the same row on desktop.
Variants	Centered (Home sections), Left-aligned with trailing link (listing pages).
States	Default; Link Hover/Focus (if a "View All" link is present).
Responsive Behaviour	Trailing link wraps below the title on mobile rather than truncating.
Accessibility	Always a real H2; never styled text pretending to be a heading.
Dependencies	Typography System.
Reusability	Every section on every listing/preview page.

Timeline

Purpose	Present chronological education/experience data as a scannable vertical narrative with a signature draw-on-scroll animation.
Responsibilities	

	Render a connecting spine and a series of timeline nodes, each with date, title, and description; draw the spine progressively as the visitor scrolls.
Inputs	Ordered array of {date, title, institution/org, description, status(optional "current" flag)}.
Variants	Alternating left/right nodes (desktop/laptop), single-side nodes (tablet), simple vertical list (mobile).
States	Default (line undrawn below the fold); Drawing (in-progress, tied to scroll offset); Node Revealed; Node Hover/Focus (expands slightly to show more detail); Current (distinct status badge).
Interactions	Scroll: continuous linear mapping of spine length to scroll offset within the section. Hover/ Focus on node: 250–350ms expand. Keyboard: nodes are sequentially focusable and readable in linear (non-visual) order regardless of alternating layout.
Responsive Behaviour	See v1.0 Section 20 per-page table — alternates spine on desktop, single-side on tablet, simple list on mobile; draw animation persists but is simplified at every breakpoint.
Accessibility	Underlying markup is a simple ordered list read top-to-bottom by assistive tech, independent of the visual alternating/graphic presentation.
Dependencies	Status Badge.
Reusability	Education page only (single instance).

Statistics Counter

Purpose	Reinforce credibility with animated numeric highlights (projects completed, years experience, etc.).
Responsibilities	Animate a number from 0 to its target value once, when first scrolled into view.
Inputs	Target numeric value, label, optional suffix ("+", "%").
States	Unrevealed (static "0"); Counting (1200–1800ms decelerating animation); Revealed (final static value).
Interactions	Scroll trigger: fires once per page view, ease-out decelerating count. Reduced motion: jumps directly to final value, no count animation.
Responsive Behaviour	Row of 3–5 desktop, wraps to 2-column grid tablet, single column mobile.
Accessibility	Final value present in the DOM as real text (not canvas/image) so screen readers announce the number even without the animation.
Reusability	Home page only.

Gallery / Carousel

Purpose	Present multiple project/blog images in a compact, browsable format.
Responsibilities	Render a scrollable/swipeable strip or grid of images; open a Modal/Lightbox on click for full-size viewing.
Inputs	Ordered array of {image, alt text, optional caption}.

Variants	Grid (desktop/laptop multi-image row), 2-up grid (tablet), single swipeable carousel (mobile).
States	Default; Hover (zoom-on-hover cue per image); Focus; Active slide (carousel mode); Loading (skeleton per image); Empty (no gallery images supplied — section omitted entirely, never rendered empty).
Interactions	Click/Tap: opens Modal/Lightbox at that image index. Touch: horizontal swipe advances carousel. Keyboard: arrow keys advance/retreat when carousel is focused; Enter opens lightbox. Mouse: optional drag-to-scroll on grid variant.
Accessibility	Carousel exposes current slide index to assistive tech (<code>aria-live="polite"</code> position announcement); all images carry descriptive alt text, not "screenshot".
Dependencies	Modal/Lightbox.
Reusability	Project Detail page only.

Modal / Lightbox

Purpose	Present a single enlarged image (or, for Certificate cards, a credential) without leaving the current page.
Responsibilities	Overlay the page with a dimmed backdrop; display the target image at maximum comfortable size; provide a close control and, in Gallery context, next/previous controls.
States	Closed; Opening (fade+scale in, 250–350ms); Open; Closing (fade+scale out); Loading (image not yet decoded — skeleton/spinner); Error (image failed — fallback message, close remains available).
Interactions	Open trigger: click/tap on Gallery image or Certificate Card. Close: click on backdrop, close (X) button, or Escape key. Keyboard: focus is trapped within the modal while open; Escape closes; Tab cycles only modal-internal controls; focus returns to the triggering element on close. Touch: swipe-down-to-close permitted as an enhancement, not required.
Accessibility	<code>role="dialog"</code> <code>aria-modal="true"</code> ; labelled by the image's alt text or a visible caption; background content marked <code>inert</code> / <code>aria-hidden</code> while open.
Reusability	Gallery (Project Detail), Certificate Card (Education).

Accordion (FAQ)

Purpose	Present question/answer pairs in a compact, progressively-disclosed format.
Responsibilities	Render a list of collapsed items; expand one item's answer on activation.
Variants	Single-open (opening a new item closes the previous one) — the required default; multi-open is not permitted, to keep the FAQ list scannable.
States	Collapsed (default); Expanded; Hover on header (background tint); Focus on header.
Interactions	Click/Tap: toggles expand/collapse with an animated height transition, 250–350ms ease-in-out. Keyboard: header is a real button, focusable, Enter/Space toggles. Icon: chevron/plus rotates to indicate state.
Accessibility	Header button carries <code>aria-expanded</code> ; answer panel carries <code>aria-hidden</code> synced to state and is programmatically associated via <code>aria-controls</code> .

Reusability	Hire Me FAQ section only.
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Tabs

Purpose	Reserved component for future content segmentation (e.g. grouping Education "Skills" by category) — not required for v1 content volume but specified for scalability (Part XII).
States	Active tab; Inactive tab; Hover; Focus; Disabled (empty tab).
Interactions	Click/Tap: switches visible panel, no page reload. Keyboard: Left/Right arrow keys move between tabs per WAI-ARIA Tabs pattern; Home/End jump to first/last tab.
Accessibility	<code>role="tablist" / role="tab" / role="tabpanel"</code> , roving tabindex.
Reusability	Not activated at launch; available for Education Skills grouping in a future revision without new component work.

Search Bar

Purpose	Let visitors live-filter Projects and Blog listings by keyword.
Responsibilities	Accept free-text input; filter the associated grid by title/tag match on every keystroke (debounced).
Inputs	Free-text query string.
Outputs	Filtered result set; empty-state trigger when no match.
States	Empty (placeholder visible); Focused; Typing (150ms debounce before filter applies); Has Query (clear "x" control appears); No Results (grid shows Empty State).
Interactions	Typing: live filter, debounced 150ms to avoid layout thrash. Clear: "x" button resets query and grid instantly. Keyboard: Escape clears the field when focused. Touch: native mobile keyboard, search-type input mode.
Accessibility	<code><input type="search"></code> with a visible, associated label (may be visually hidden but must exist); filtered result count announced via <code>aria-live="polite"</code> .
Dependencies	Filter Chip, Empty State.
Reusability	Projects listing, Blog listing.

Tag / Badge / Filter Chip

Purpose	Compactly label content by category/technology (Tag), communicate status (Badge — e.g. "Current"), or act as a selectable filter control (Filter Chip).
Variants	Static Tag (non-interactive, on cards), Status Badge (non-interactive, colored per Design Tokens status colors), Filter Chip (interactive, toggleable).
States	Tag: Default only. Badge: Default (per status color). Filter Chip: Unselected, Selected (accent fill), Hover, Focus, Disabled (category with zero matching items — reduced opacity, non-interactive).
Interactions	Filter Chip — Click/Tap: toggles selected state, re-filters associated grid with a 250–350ms fade. Keyboard: focusable, Enter/Space toggles. Static Tag/Badge — no interaction.

Accessibility	Filter Chip uses <code>aria-pressed</code> to expose toggle state; Status Badge text is never conveyed by color alone (icon or text label always present).
Reusability	Tags: all cards. Status Badge: Education "Current" indicator. Filter Chip: Projects and Blog listing filter bars.

Toast (Notification)

Purpose	Confirm a lightweight, non-blocking action (e.g. "Link copied") without interrupting the page.
Responsibilities	Appear briefly near the triggering control or a fixed corner; auto-dismiss.
States	Entering (fade+slide in, 200–250ms); Visible (holds 2–3s); Exiting (fade out, 200ms); Success/Error variants (icon + tone differ per Design Tokens status colors).
Interactions	Trigger: Blog Detail "copy link" share action. Dismiss: auto after timeout; manual dismiss not required but permitted via click.
Accessibility	<code>role="status"</code> <code>aria-live="polite"</code> so assistive tech announces it without moving focus.
Reusability	Blog Detail share action; reserved for future use (e.g. Contact form success where a persistent inline message is not used instead).

20.5 Form Components — Form, Input Field, Dropdown, Contact Form

Field	Specification
Purpose	Collect and validate visitor input (Contact form: Name, Email, Subject, Message).
Responsibilities	Render labeled fields; validate on blur and on submit; show inline errors; manage submit loading/success/failure states; prevent duplicate submission.
Inputs	Field values as typed; field configuration (type, required, validation pattern).
Outputs	Validated payload on successful submit; per-field error messages on failure.
Variants	Text Input (Name), Email Input (Email, with format validation), Select/Dropdown (Subject/Reason), Textarea (Message).
States (per field)	Empty; Focused (label animates upward per v1.0 Section 16 spec); Filled; Valid (subtle success cue on blur, only after first interaction — never validate before the visitor has touched the field); Invalid (border + icon + inline message in <code>--color-danger</code>); Disabled (during submit-in-flight).
States (form-level)	Idle; Submitting (submit button shows Loading state per Button spec, all fields disabled); Success (form replaced by a confirmation message with a checkmark motion per v1.0 Section 18); Failure (inline error banner above the fields, submitted values preserved, fields re-enabled).
Interactions	Focus: label animates upward, border/underline shifts to accent. Blur: field validates; error appears if invalid. Typing: error clears live once the field becomes valid, without waiting for the next blur. Keyboard: Tab moves field-to-field in visual order; dropdown is operable via arrow keys + Enter. Touch: appropriate mobile keyboard type per field (email keyboard for Email field).
Accessibility	Every field has a real, programmatically associated <code><label></code> ; errors use <code>aria-describedby</code> linking the field to its error text and <code>aria-invalid="true"</code> ; color is never the sole error

	indicator (icon + text always present); the Select/Dropdown is a native <code><select></code> or a fully ARIA-compliant custom listbox, not a non-semantic styled div.
Dependencies	Primary Button (submit), Toast or inline success message.
Reusability	Contact page (primary instance). The Search Bar (Section 20.4) is a related but distinct single-field component, not a Form variant.

20.6 Feedback & Loading Components

Loader (Page Loader)

Purpose	Mask initial asset/font loading with a branded, premium moment before content appears.
States	Loading (animated mark/progress indicator); Complete (fades out into loaded page, 600–1000ms).
Interactions	No user interaction; automatic on initial page load only, once per session per page.
Accessibility	<code>aria-busy="true"</code> on the document while active; must not block screen-reader access if it exceeds 3 seconds (fallback: skip loader).
Reusability	Every primary page, first load only.

Skeleton Loader

Purpose	Represent the shape of not-yet-loaded card/gallery content, preventing layout shift (supports CLS budget, Part XI).
Responsibilities	Render a placeholder matching the exact dimensions of the eventual Card/Gallery image, with a subtle shimmer animation.
States	Loading (shimmer, looping ease-in-out); Resolved (crossfades to real content, 200–300ms).
Accessibility	Marked <code>aria-hidden="true"</code> and excluded from the accessibility tree while loading; replaced content is announced normally once resolved.
Reusability	Below-the-fold images across Projects, Blog, Education (Certificates), Project Detail Gallery — anywhere lazy-loading is used (Part XI).

20.7 Identity Components

Social Links

Purpose	Provide consistent, always-available links to LinkedIn, GitHub, Instagram, WhatsApp/Email.
States	Default; Hover (icon fill/color shift + micro-scale); Focus.
Interactions	Click/Tap: opens the relevant profile/chat in a new tab (all except Email, which opens a <code>mailto:</code> link) or the platform's native app on mobile where supported.
Accessibility	Each icon-only link carries a descriptive <code>aria-label</code> (e.g. "Ammlan Rath on LinkedIn, opens in new tab").
Reusability	Home, Footer (every page), Contact page (expanded with Location/Business Hours).

Component Count Summary: 35 named reusable components/variants fully specified above, organized into 7 families for consistency (Navigation, Buttons, Cards, Content Primitives, Interactive Overlays, Forms, Feedback/Identity). Every component listed in the original inventory request (Navbar, Footer, Hero, Section Header, Primary/Secondary/Outline Button, CTA, Project/Blog/Certificate/Achievement Card, Timeline, Skill Card, Statistics Counter, Gallery, Carousel, Modal, Accordion, Tabs, Search Bar, Tag, Badge, Breadcrumb, Toast, Forms, Input Fields, Dropdown, Loader, Skeleton Loader, Back To Top Button, Social Links, Profile Card, Image Card, Contact Form) is covered above, either individually or within a documented family group that preserves full per-variant differentiation.

Part IV — Page-by-Page Functional Breakdown & Wireframes

Every page is documented against a fixed structure: Purpose/Objectives/Audience (summary; full detail in v1.0 Section 18 and Part I), Structural Wireframe (Desktop/Laptop/Tablet/Mobile), Component Hierarchy, Section Ordering & Content Flow, Functional Requirements, Navigation Behaviour, SEO, Accessibility, Responsive Behaviour, Performance Considerations, Future Scalability, and Acceptance Criteria. Wireframes are structural (box/region descriptions only) — no visual mockups, per stakeholder instruction.

21.1 Home Page

Structural Wireframe

Breakpoint	Layout Description
Desktop (1440px+)	Full-viewport Hero (headline centered, 2 CTAs inline below, scroll-hint bottom-center) → About Preview (2-col: portrait left, text+link right) → Skills Preview (8–12 icon+label items, 6–8 col grid) → Featured Projects (3–4 col card grid + "View All" trailing link) → Featured Blogs (3 col card row + "Read All" link) → Statistics (5-col counter row) → Resume Download band (full-width, centered content) → Freelance CTA band (full-width, high-contrast, centered) → Social Links row → Footer (multi-column).
Laptop (1024–1439px)	Identical section order; grids condense one column tighter (Projects 3-col, Skills 6-col); Hero height slightly reduced.
Tablet (768–1023px)	Same section order; About Preview stacks to single column with portrait above text; Skills/Projects/Blogs grids reduce to 2 columns; Statistics wraps to 2-column grid.
Mobile (320–767px)	All sections stack single-column, full width; Hero height capped to reduce scroll-before-content; CTA buttons stack vertically full-width; Statistics single column; Footer collapses to one stacked column.

Component Hierarchy

Navbar → Hero(Primary Button, Secondary Button) → SectionHeader+AboutPreview → SectionHeader+SkillCard×N → SectionHeader+ProjectCard×3-4 → SectionHeader+BlogCard×2-3 → StatisticsCounter×3-5 → CTA(Resume) → CTA(Freelance) → SocialLinks → Footer(SocialLinks, nav list).

Section Ordering & Content Flow

Hero → About Preview → Skills Preview → Featured Projects → Featured Blogs → Statistics → Resume Download → Freelance CTA → Social Links → Footer. This order is fixed and intentional: identity first, proof of skill second, proof of work third, thought leadership fourth, credibility signals fifth, then two conversion opportunities before exit. Reordering any of these ten sections requires a change request (Appendix C process).

Functional Requirements (Behavioural)

Element	Trigger	Expected Behaviour	Error/Edge Handling
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"View My Work" button	Click/Tap/Enter	Navigates to Projects listing page	N/A — internal route, cannot fail
"Hire Me" button	Click/Tap/Enter	Navigates to Hire Me page	N/A
Featured Project Card	Click/Tap on card	Navigates to that project's Project Detail page	If the target file is missing, resolves to the 404 page (FR-12, zero dead links at launch)
Statistics Counter	Scroll into view	Animates 0 → target once	If JS fails to load, final value must still render statically (progressive enhancement)
Resume Download band CTA	Click/Tap	Opens/downloads the resume PDF from <code>/assets/resume/</code> in a new tab	If the asset is missing, the link must be visibly disabled at build time rather than 404 in production
Freelance CTA band button	Click/Tap	Navigates to Hire Me page	N/A

Navigation, SEO, Accessibility, Responsive, Performance, Scalability, Acceptance

Navigation, SEO metadata, and accessibility requirements for this page are unchanged and binding from v1.0 Sections 18.1.10–18.1.13; responsive behaviour is defined in the wireframe table above and v1.0 Section 20; performance targets are the site-wide budget in Part XI, applied with extra weight given this page loads first. Future scalability: Featured Projects/Blogs remain a fixed curated set (Part XII).

Acceptance Criteria — Home Page (all must pass): (1) H1 and meta title both state owner name + role. (2) All ten sections render in the fixed order above at all four breakpoints with no horizontal scroll. (3) Resume link resolves to a real, downloadable file. (4) Statistics display correct final values with JS disabled. (5) All Featured Project/Blog cards route to a live, non-404 detail page. (6) LCP ≤2.5s on simulated mid-range mobile (Part XI). (7) Every interactive element passes keyboard-only operation. (8) No animation exceeds the Part II Motion System duration ceiling.

21.2 About Page

Structural Wireframe

Breakpoint	Layout Description
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Desktop/ Laptop	Compact Hero (page title) → single-column narrative body with one full-width portrait/image moment and 1–2 full-width pull-quotes interrupting the column → Core Values row (3–5 small cards) → Interests section (text or icon+label list) → Personality pull-quote → Technical Growth paragraph bridging to Education → Closing CTA row ("See My Journey" / "See My Work") → Footer.
Tablet	Same order; portrait moves above the text block it accompanies, full width.
Mobile	Fully stacked single column; pull-quotes remain full-width but reduce font size per Typography scale rules; Core Values cards stack single column.

Component Hierarchy

Navbar → Hero(compact) → PullQuote → NarrativeBody → PullQuote → ValueCard×3-5 → NarrativeBody(Interests) → PullQuote(Personality) → NarrativeBody(Technical Growth) → SecondaryButton×2 → Footer.

Functional Requirements

Element	Trigger	Expected Behaviour
"See My Journey" button	Click/Tap	Navigates to Education page
"See My Work" button	Click/Tap	Navigates to Projects page
Inline portrait image	Hover (desktop only)	Subtle zoom-on-hover per Card Hover-family timing; no click behaviour

Acceptance Criteria — About Page: (1) Strict heading hierarchy, no skipped levels. (2) All required content sections present (Journey, Vision, Mission, Values, Interests, Personality, Technical Growth). (3) Both closing CTAs route correctly. (4) All imagery has descriptive alt text. (5) Body text meets 4.5:1 contrast. (6) Page readable and fully functional with motion disabled.

21.3 Education Page

Structural Wireframe

Breakpoint	Layout Description
Desktop/ Laptop	Compact Hero → vertical Timeline (alternating left/right nodes around a center spine) → Semester Results (compact table/chart) → Certificates grid (3–4 col) → Achievements list/grid → Current Experience (status-badged entry, may be embedded in Timeline or its own band) → Workshops list → Skills grid (grouped by category) → Future Goals paragraph → soft CTA → Footer.
Tablet	Timeline shifts to single-side layout, spine left-aligned; Certificates grid 2-col.
Mobile	Timeline becomes a simple vertical list with compact node markers, draw animation simplified; Certificates/Skills grids single column.

Component Hierarchy

Navbar → Hero(compact) → Timeline(node×N, StatusBadge on current) → Table(SemesterResults) →
 SectionHeader+CertificateCard×N → SectionHeader+AchievementCard×N →
 SectionHeader+SkillCard×N(grouped) → NarrativeBody(FutureGoals) → SecondaryButton → Footer.

Functional Requirements

Element	Trigger	Expected Behaviour	Edge Case
Timeline node	Hover/Focus	Expands to reveal additional detail (250–350ms)	On touch devices, tap expands in place rather than requiring hover
Certificate Card	Click/Tap	Opens credential image in Modal/Lightbox, or opens an external verification URL in a new tab if provided	If neither image nor link exists, card renders in Disabled state (no click target) rather than being omitted, to preserve credential visibility
"Current" status badge	Static	Visually and semantically marks one active entry	At most one entry should carry "Current" at a time per program; multiple concurrent current entries (e.g. degree + internship) are permitted if both are genuinely ongoing

Acceptance Criteria — Education Page: (1) Timeline content is readable in linear order by screen readers regardless of visual alternating layout. (2) Every certificate has a title, issuer, and date at minimum. (3) Timeline draw animation respects reduced-motion. (4) Grids reflow correctly at all four breakpoints with no overlap. (5) No dead links among certificate verification URLs.

21.4 Projects Page (Listing)

Structural Wireframe

Breakpoint	Layout Description
Desktop/ Laptop	Compact Hero → sticky/top-anchored Filter bar (Category chips + Technology chips + Search input, horizontal) → 3–4 col Project Card grid → Empty State (conditional) → page-level "Didn't find what you need? Hire Me" CTA band → Footer.
Tablet	Filter bar horizontal, may wrap to two rows; grid 2-col.
Mobile	Filter bar collapses into a dropdown/expandable panel triggered by a "Filters" button; grid single column.

Component Hierarchy

Navbar → Hero(compact) → SearchBar + FilterChip×N → ProjectCard×N (or EmptyState) → CTA band → Footer.

Functional Requirements

Element	Trigger	Expected Behaviour	Error/Edge Handling
Filter Chip	Click/Tap	Toggles selected; grid re-filters instantly with 250–350ms fade; multiple chips combine as AND across categories, OR within a category	Selecting a combination with zero matches shows Empty State with "Clear Filters" action
Search input	Typing (debounced 150ms)	Live-filters grid by title/tag substring match, combined with any active chip filters	Empty query restores the chip-filtered (or full) set
Project Card	Click/Tap anywhere on card	Navigates to that project's Project Detail page	Broken target resolves to 404 (must not occur at launch per FR-12)
"Clear Filters" (Empty State)	Click/Tap	Resets all chips and search query, restores full grid	—

Acceptance Criteria — Projects Page: (1) Filtering never triggers a full page reload. (2) Filtered result count is announced to assistive tech via a live region. (3) Every project card routes to a live detail page. (4) Filter/search state does not persist unexpectedly across a fresh page load (resets on navigation away and back, unless a future URL-query-param enhancement is implemented — Appendix C). (5) Grid reflows correctly at all breakpoints with correct column counts per the responsive table.

21.5 Project Detail Page

Structural Wireframe

Breakpoint	Layout Description
Desktop/ Laptop	"Back to Projects" link → Hero (title, tagline, category, hero image, Visit Live Site + View Source buttons) → Gallery (multi-image row) → Overview → Problem Statement → Features (bulleted) → Challenges/Solutions (paired blocks) → Tech Stack chips → Development Process → Future Improvements → outbound Live/Source buttons (repeat) → "Hire Me for Something Similar" CTA → Related Projects (2–3 cards) → Footer.
Tablet	Gallery becomes a 2-up grid or swipeable row; single-column body content otherwise unchanged.
Mobile	Gallery becomes a single swipeable carousel; Challenge/Solution pairs stack vertically (challenge then its solution, never side-by-side).

Component Hierarchy

Navbar → Breadcrumb → Hero(detail variant, PrimaryButton×2) → Gallery/Carousel → NarrativeBody×N → TechStackChip×N → PrimaryButton(CTA) → ProjectCard×2-3(Related) → Footer.

Functional Requirements

Element	Trigger	Expected Behaviour	Edge Case
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"Visit Live Site"	Click/Tap	Opens the project's live URL in a new tab (<code>target="_blank"</code> <code>rel="noopener"</code>)	If no live URL exists, button is omitted (never rendered disabled) — "View Source" becomes the sole primary CTA
"View Source Code"	Click/Tap	Opens the repository URL in a new tab	Same omission rule as above if unavailable
Gallery image	Click/Tap	Opens Modal/Lightbox at that index; arrow keys/swipe navigate between images while open	Single-image galleries omit next/prev controls
Related Project Card	Click/Tap	Navigates to that project's own detail page	Related set is computed by shared category/technology; if fewer than 2 matches exist, show whatever matches (0–2), never an empty broken layout

Acceptance Criteria — Project Detail: (1) Every required content section (Section 18.5.3 of v1.0) is present for every project at launch. (2) Both outbound buttons, where present, open in a new tab with `rel="noopener"`. (3) Gallery/Lightbox is fully keyboard-operable. (4) Meta title/description are unique per project. (5) "Back to Projects" link is present above the hero at every breakpoint.

21.6 Blog Page (Listing)

Structural wireframe, component hierarchy, and functional requirements mirror the Projects Page pattern above (Section 21.4) with these differences: a single Featured Blog card is pinned above the grid at all breakpoints (full-width desktop/tablet, visually distinguished by size/border on mobile rather than width); cards display reading time and publish date in addition to category and tags; no "Hire Me" CTA band is required at page end.

Functional Requirements (deltas from Projects Page pattern)

Element	Trigger	Expected Behaviour
Featured Blog card	Click/Tap	Navigates to that post's Blog Detail page — identical behaviour to a regular Blog Card, larger presentation only
Category filter / Search	Same as Projects	Identical interaction pattern for cross-site learnability (v1.0 requirement)

Acceptance Criteria — Blog Page: Same criteria as Projects Page (Section 21.4), plus: Featured Blog is always exactly one post, determined by most-recent-publish-date unless manually overridden in the content log (Appendix D).

21.7 Blog Detail Page

Structural Wireframe

Breakpoint	Layout Description
Desktop/ Laptop	"Back to Blog" link → Article Hero (cover image, title, meta row: author/date/category/reading time) → Share buttons row → centered narrow-measure article body (headed sections, inline images with captions, code blocks where relevant) → Related Blogs (2–3 cards) → Previous/Next Post control → Footer.
Tablet	Article column widens proportionally, margins reduce.
Mobile	Article column becomes near-full-width with minimal side padding, preserving readable line length per Typography rules.

Functional Requirements

Element	Trigger	Expected Behaviour	Edge Case
Share — Copy Link	Click/Tap	Copies the canonical post URL to clipboard; shows Toast confirmation	If clipboard API is unavailable, falls back to a visible, selectable URL field
Share — LinkedIn/X	Click/Tap	Opens the platform's share intent URL in a new tab, pre-filled with post title + link	—
Code block — Copy	Click/Tap	Copies code content to clipboard, brief inline confirmation	—
Previous/Next control	Click/Tap	Navigates to the chronologically adjacent post	First post has no "Previous"; latest post has no "Next" — control omits the unavailable side rather than disabling it
Related Blog Card	Click/Tap	Navigates to that post's Blog Detail page	Computed by shared category/tag, same 0–3 fallback rule as Related Projects

Acceptance Criteria — Blog Detail: (1) Strict semantic heading order within the article body. (2) Meta title/description/keywords unique per post; Article schema present. (3) Reading-progress indicator (if implemented) respects reduced-motion. (4) All code blocks announce their language to assistive tech. (5) Previous/Next correctly reflects publish-date order at all times, including after new posts are added.

21.8 Hire Me Page

Structural Wireframe

Breakpoint	Layout Description
------------	--------------------

Desktop/ Laptop	Hero → Services grid (3-col) → Workflow (horizontal numbered step diagram) → Pricing Card row → Client Expectations block → FAQ Accordion → closing Inquiry Process section with Primary Button to Contact → Footer.
Tablet	Services grid 2-col; Workflow may wrap to two rows.
Mobile	Services grid single column; Workflow becomes a vertical step list.

Functional Requirements

Element	Trigger	Expected Behaviour
Workflow step	Scroll into view	Steps reveal in sequence, staggered, as the section enters the viewport
FAQ Accordion item	Click/Tap	Expands answer, collapses any previously open item (single-open pattern, Section 20.4)
"Start a Project" / "Get in Touch" button (all instances)	Click/Tap	Navigates to Contact page

Acceptance Criteria — Hire Me: (1) Pricing is always labeled as an estimate ("Starting from..."), never presented as a fixed quote. (2) Every FAQ item has visible expand/collapse state exposed to assistive tech. (3) Workflow steps are readable in linear document order regardless of horizontal/vertical visual presentation. (4) All primary CTAs route to Contact without exception.

21.9 Contact Page

Structural Wireframe

Breakpoint	Layout Description
Desktop/ Laptop	Compact Hero → Two-column layout: Contact Form (left) Direct Channels + Location + Business Hours (right, Info Card style) → Footer.
Tablet	Two-column retained if width allows, otherwise stacks (form first).
Mobile	Single column, Contact Form first, then direct channels/location/hours below it.

Functional Requirements

Element	Trigger	Expected Behaviour	Error/Edge Handling
Contact Form submit	Click/Tap/ Enter on Submit	Validates all fields; on success, shows Success state (form replaced by confirmation message with checkmark motion); button shows Loading state during submission	On failure, form shows an inline error banner, preserves entered values, re-enables fields; network/timeout errors display a generic retry message, never a raw error code
	Click/Tap	Opens the respective channel in a new tab or native app	—

WhatsApp / Instagram / LinkedIn / GitHub icon links			
Email link	Click/Tap	Opens default mail client via mailto;; plain-text address also shown for manual copy	—

Acceptance Criteria — Contact Page: (1) All form fields have associated labels and inline validation. (2) Success and failure states are both implemented and visually distinct without relying on color alone. (3) No precise street address is shown (city/region only, per privacy requirement). (4) Business hours/response- time expectation is visible without scrolling on desktop.

21.10 Error & Status Pages (400 / 401 / 403 / 404 / 500 / Maintenance / Offline)

Structural Wireframe (shared pattern, all seven pages)

Breakpoint	Layout Description
All breakpoints	Centered composition: simplified header (logo only) → large stylized status indicator → short headline → 1–2 sentence plain-language explanation → 1–2 recovery action buttons → full footer navigation. Composition remains centered at every breakpoint; only surrounding whitespace and button stacking (vertical, full-width on mobile) change.

Functional Requirements

Page	Recovery Actions	Behaviour
400	Go to Home, Contact Support	Home routes to "/"; Contact Support routes to Contact page
401	Go to Home, Contact Owner	Same pattern; reserved for future access-controlled areas (Appendix B assumption)
403	Go to Home	Single recovery action only
404	Go to Home, View Projects, Search	"Search" scrolls to or focuses a lightweight search entry point if present, otherwise routes to Projects with Search Bar pre-focused
500	Go to Home, Try Again	"Try Again" re-requests the previous route
Maintenance	Social links only (no functional recovery action required)	Static, informational
Offline	Retry	Re-attempts the network request; minimally cached content shown where feasible

Acceptance Criteria — Error Pages: (1) All seven pages share one stylesheet/script (`error.css` / `error.js`). (2) All are set `noindex` and excluded from `sitemap.xml` . (3) Status/headline is announced

immediately to screen readers. (4) Global footer navigation remains present and functional on every error page. (5) No error page ever renders a raw stack trace or raw HTTP status dump.

Part V — Complete User Flow Documentation

Every important user journey is documented below with Entry Point, Goals, Navigation Path, Decision Points, Primary Actions, Secondary Actions, and Exit Points, for each of the six audience types defined in v1.0 Section 6.

22.1 Recruiter Flow

Entry Point	Home (direct link from resume/LinkedIn) or Projects (shared project link)
Goals	Assess credibility and skill level in under two minutes
Navigation Path	Home Hero → About Preview scan → Skills Preview scan → Featured Projects scan → (branch) Projects listing for deeper look → (branch) Education for credential check → Resume Download
Decision Points	(1) Is the first-viewport identity/role clear? If not, likely exit within 5s. (2) Does a Featured Project look credible? If yes, proceeds to Project Detail; if no, checks Education for credentials instead.
Primary Actions	Download Resume; open a Project Detail page
Secondary Actions	Open Education page; open LinkedIn/GitHub via Social Links
Exit Points	Resume download (success exit); LinkedIn/GitHub outbound (delegated exit); browser back/tab close (abandonment, most common if Hero fails Decision Point 1)

22.2 Freelancer Client Flow

Entry Point	Home (referral/search) or Hire Me (direct link shared by owner)
Goals	Judge design taste, then understand pricing/process, then decide whether to reach out
Navigation Path	Home Hero/scroll for taste → Freelance CTA band → Hire Me (Services → Workflow → Pricing → FAQ) → Contact
Decision Points	(1) Does the visual quality on Home feel "premium" enough to trust? (2) Do the listed Services match the client's need? (3) Does the Pricing placeholder fall within an acceptable range? (4) Does the FAQ resolve remaining doubts (ownership, timeline, revisions)?
Primary Actions	Submit Contact form with project details
Secondary Actions	Message via WhatsApp for a faster, informal first contact; browse Projects for relevant prior work
Exit Points	Contact form success confirmation (success exit); WhatsApp outbound (delegated success exit); abandonment at Pricing if expectations mismatch

22.3 Company / Hiring Team Flow

Entry Point	Projects listing (shared by an internal recruiter) or Home
Goals	Evaluate technical depth and communication ability across multiple pieces of work
Navigation Path	

	Projects listing → filter by relevant technology → open 2–3 Project Detail pages → Education (credential cross-check) → Contact or external follow-up
Decision Points	(1) Do the Challenges/Solutions sections reveal genuine problem-solving? (2) Is the Tech Stack aligned with the role's requirements? (3) Is there enough evidence across ≥2 projects to proceed to a live interview?
Primary Actions	Open multiple Project Detail pages; use Filter/Search to narrow by required technology
Secondary Actions	Open Source Code links; check Education for formal credentials
Exit Points	Source Code outbound (delegated exit for deeper technical review); Contact form (direct exit); abandonment if fewer than 2 projects meet the technology filter

22.4 Student Flow

Entry Point	Blog (search result or shared post link) or Home
Goals	Find inspiration and practical guidance; benchmark own progress
Navigation Path	Blog listing → filter/search by topic → Blog Detail (full read) → Related Blogs or Previous/Next → Education (benchmark comparison) → About (relatable journey narrative)
Decision Points	(1) Does the Featured Blog or a filtered result match current interest/skill level? (2) Is the reading time acceptable given available time? (3) After reading, is there enough momentum to read another post or check Education?
Primary Actions	Read a Blog Detail article end-to-end
Secondary Actions	Navigate Previous/Next; open Education to compare trajectory; open About for narrative context
Exit Points	Natural end-of-article (soft exit, often continues via Related/Previous-Next); browser back if the post doesn't match expectations early

22.5 Developer / Peer Flow

Entry Point	Project Detail (shared link) or Projects listing
Goals	Judge real technical depth, not just visual polish
Navigation Path	Project Detail → Tech Stack → Challenges/Solutions → Development Process → Future Improvements → Source Code (outbound) → Related Projects (repeat cycle)
Decision Points	(1) Does the Challenges section name real, specific engineering problems (not generic filler)? (2) Does the Source Code link work and reflect the described stack? (3) Is Future Improvements reflective of genuine engineering judgment?
Primary Actions	Open Source Code in a new tab
Secondary Actions	Open additional Related Projects; read a technical Blog post if available
Exit Points	Source Code outbound (primary, delegated exit to GitHub); abandonment if Challenges/Solutions reads as generic

22.6 General Visitor Flow

Entry Point	Home (social share, casual link)
Goals	Browse casually, be impressed by design and motion
Navigation Path	Home — full linear scroll through all ten sections, no strong branch intent
Decision Points	(1) Is the motion/visual quality engaging enough to keep scrolling past the Hero? (2) Does any single section (a project, a stat) prompt a deeper click out of curiosity?
Primary Actions	Scroll through Home end-to-end
Secondary Actions	Occasional opportunistic click into a Project or Blog card
Exit Points	Footer (natural end-of-scroll exit); early abandonment if motion/visual quality underwhelms within the first 1–2 sections

Part VI — Content Guidelines

23.1 Tone of Voice & Writing Style

Attribute	Rule
Overall Tone	Confident, precise, understated — never boastful, never apologetic. Describes work in terms of problems solved and decisions made, not adjectives ("amazing", "revolutionary").
Person	First person ("I") throughout About, Project Detail process notes, and Blog. Third person is never used for the owner.
Sentence Length	Favor short-to-medium sentences (12–20 words average). Long compound sentences are broken up, especially in Project Detail and Blog Detail body copy.
Technical Register	Precise and specific (name the actual technology/technique) rather than vague ("used modern tools").

23.2 Heading & Paragraph Style

Element	Rule
Headings	Sentence case, not Title Case (e.g. "Why I build what I build", not "Why I Build What I Build"). Verb-first or noun-first, never a question mark unless genuinely posing a question to the reader.
Paragraphs	3–5 sentences maximum per paragraph in long-form content (About, Blog Detail); shorter (1–3 sentences) in card descriptions and previews.

23.3 CTA Wording & Button Text Standards

Context	Standard
All buttons	Verb-first, 1–3 words ("View Project", "Download Resume", "Start a Project"). Never "Click Here" or "Submit" alone.
Primary conversion CTAs	Action + outcome implied ("Hire Me", "Start a Project") rather than passive ("Learn More about Hiring").
Secondary/soft CTAs	Lower-commitment verbs ("Read More", "See My Journey").

23.4 Error, Success & Empty State Messages

Message Type	Tone Rule	Example Pattern (not verbatim copy)
Form field error	Specific, non-technical, tells the visitor exactly what to fix	"[Field] is required" / "Enter a valid email address"
Form submit success	Warm, sets expectation for what happens next	Confirms receipt and states the expected response window (matches Contact page Business Hours)

Form submit failure	Reassuring, offers a clear next step, never a raw error code	Apologizes briefly, suggests retry or an alternative channel (e.g. email/WhatsApp)
Empty state (filters/search)	Light, non-blaming, action-oriented	States nothing matched and offers a "Clear Filters" action
404 page	Light, slightly playful, on-brand — the one place a little personality in error copy is encouraged	Softens the dead-end without being flippant about it
500 / Offline	Reassuring, takes ownership (500) or clarifies it's a network issue (Offline), never blames the visitor	—

23.5 Blog Writing Guidelines

Element	Rule
Opening paragraph	States the reader's takeaway within the first 2 sentences — never a slow narrative wind-up.
Structure	Broken into clearly headed H2/H3 sections; no single unbroken block exceeding ~150 words.
Code examples	Always labeled with language; accompanied by 1–2 sentences of explanation, never dropped in unexplained.
Length	No fixed minimum/maximum; reading time is calculated and displayed, not manually estimated.
Categories	Every post assigned exactly one primary category from the fixed taxonomy (Section 18.6.3 of v1.0); additional topical tags (1–3) may cross categories.

23.6 Project Description Guidelines

Element	Rule
One-line description (card)	States what the project is and who it's for in one sentence, never a feature list.
Problem Statement	Written from the user/business problem's perspective, not "I wanted to practice X technology" (technology practice is a valid private motivation but not the stated problem).
Challenges/Solutions	Always paired 1:1; each Challenge names a specific technical or product obstacle, each Solution explains the actual resolution, not just "I fixed it."
Future Improvements	At least one genuine, specific item per project — signals reflective engineering judgment (v1.0 requirement); never left as a placeholder.

Part VII — Image Specifications

Image Type	Recommended Dimensions	Aspect Ratio	Format	Loading	Naming Convention
Profile / Portrait	800×1000px min	4:5	WebP (JPEG fallback)	Eager (Home/ About hero-adjacent)	profile-[context].webp
Hero Background (Home)	2560×1440px	16:9	WebP, aggressively compressed	Eager, highest priority	hero-home-bg.webp
Project Hero Image	1600×900px	16:9	WebP	Eager (above fold on Project Detail)	[project-slug]-hero.webp
Project Thumbnail (card)	640×400px	16:10	WebP	Lazy	[project-slug]-thumb.webp
Project Screenshot (gallery)	1440×900px	16:10	WebP	Lazy, except first gallery image	[project-slug]-shot-01.webp (sequential)
Blog Banner / Cover	1200×630px	1.91:1 (matches OG default)	WebP	Eager on detail page, lazy on listing cards	[post-slug]-cover.webp
Certificate Image	1000×750px	4:3	WebP or PNG (if credential has transparency/ text needing lossless clarity)	Lazy	cert-[issuer]-[year].webp
Achievement Image (optional)	800×600px	4:3	WebP	Lazy	achievement-[slug].webp
Icons (UI/Skill)	24×24px – 48×48px	1:1	SVG (preferred) or icon font	Eager, inlined where possible	icon-[name].svg
Favicon	512×512px source,	1:1	ICO/PNG set	Eager (browser-requested)	favicon.ico at root

	exported as multi-size .ico				
Open Graph Image	1200×630px	1.91:1	JPEG or PNG (per platform compatibility)	N/A — fetched by crawlers, not rendered in-page	og - [page-slug].jpg

Rule	Specification
Compression	All raster images compressed to the smallest file size that shows no visible quality loss at the largest breakpoint they appear at; target <200KB for hero images, <80KB for cards, <40KB for thumbnails.
Lazy Loading	Every below-the-fold image uses native <code>loading="lazy"</code> ; only each page's immediate hero-visible asset(s) load eagerly.
Responsive Behaviour	All images use <code>srcset / sizes</code> to serve an appropriately downscaled asset per breakpoint rather than shipping one oversized file to mobile.
Layout Stability	Every image declares explicit <code>width / height</code> (or an aspect-ratio CSS box) to prevent Cumulative Layout Shift (Part XI).

Part VIII — Accessibility Implementation Guide

Baseline target: WCAG 2.1 Level AA, site-wide, with no exceptions. This part expands v1.0 Section 20/21 into an implementation-level checklist.

Area	Implementation Requirement
Keyboard Navigation	Every interactive element (links, buttons, form fields, filters, accordions, gallery/lightbox controls, carousel) reachable and fully operable using Tab, Shift+Tab, Enter, Space, and Arrow keys alone, matching each component's spec in Part III. Tab order follows visual/reading order on every page.
Screen Reader Support	Semantic landmark elements (<code>header</code> , <code>nav</code> , <code>main</code> , <code>footer</code> , <code>article</code> , <code>section</code>) on every page; heading levels strictly non-skipping; dynamic content changes (filtered grids, form errors, toast) announced via <code>aria-live</code> regions per Part III component specs.
Focus Management	Visible focus ring (not color-only) on every focusable element, using <code>--color-focus</code> token; modals/lightboxes trap focus while open and return focus to the trigger on close; mobile nav overlay traps focus while open.
Colour Contrast	Body text $\geq 4.5:1$ against its background; large headline text ($\geq 24\text{px}$ bold or $\geq 18.5\text{px}$) may use the 3:1 allowance but should exceed it given the dark theme; UI component boundaries (focus rings, form field borders) $\geq 3:1$.
Alt Text Standards	Every meaningful image has descriptive alt text describing what is shown and its relevance (never "screenshot" or "image"); purely decorative graphics use empty <code>alt=""</code> so they are skipped correctly.
Accessible Forms	Every field has a real, programmatically associated <code><label></code> ; errors linked via <code>aria-describedby</code> and flagged with <code>aria-invalid</code> ; color never the sole error/success signal.
Accessible Navigation	A "skip to main content" mechanism available on every page load; current page marked <code>aria-current="page"</code> in the Navbar.
Accessible Buttons	Real <code><button></code> / <code><a></code> elements as semantically appropriate; icon-only buttons carry descriptive <code>aria-label</code> .
Responsive Accessibility	Touch targets minimum $44 \times 44\text{px}$ at all breakpoints; zoom/pinch never disabled via viewport meta; reflow supported up to 400% zoom without horizontal scrolling of the page body.
Reduced Motion	Every Part II/VI animation is suppressed or reduced when <code>prefers-reduced-motion: reduce</code> is set; content remains fully present and usable.
ARIA Usage Rule	ARIA roles/states used only where semantic HTML is insufficient (e.g. <code>aria-expanded</code> on accordions, <code>aria-pressed</code> on filter chips) — never as a substitute for correct native elements.

Part IX — SEO Requirements Per Page (Expanded)

The per-page metadata pattern table from v1.0 Section 21 remains fully binding (Meta Title Pattern, Meta Description Focus, and Notes for Home through Error pages). This part adds the remaining required fields per page: URL Slug, Canonical URL, and Robots Indexing Rule.

Page	URL Slug	Canonical URL Pattern	Robots Rule
Home	/	https://[domain]/	index, follow
About	/html/about.html	Self-referencing canonical	index, follow
Education	/html/education.html	Self-referencing canonical	index, follow
Projects	/html/project.html	Self-referencing canonical	index, follow
Project Detail (each)	/html/projects/[project-slug].html	Self-referencing canonical (prevents duplicate-content ambiguity across template reuse)	index, follow
Blog	/html/blog.html	Self-referencing canonical	index, follow
Blog Detail (each)	/html/blogs/[post-slug].html	Self-referencing canonical	index, follow
Hire Me	/html/hire.html	Self-referencing canonical	index, follow
Contact	/html/contact.html	Self-referencing canonical	index, follow
All Error/Status Pages	/html/errors/[code].html	N/A	noindex, nofollow; excluded from sitemap.xml

Open Graph and Twitter Card metadata (og:title, og:description, og:image, og:url, summary_large_image) apply per-page exactly as defined in v1.0 Section 21.1, with the image asset for each page drawn from the Open Graph Image row in Part VII. Structured data recommendations (Person schema on Home/About, Article schema on Blog Detail, CreativeWork schema on Project Detail) remain binding recommendations.

Part X — Performance Budget

Metric	Target
Largest Contentful Paint (LCP)	< 2.5s on simulated mid-range mobile connection (4G, ~1.6Mbps down)
First Contentful Paint (FCP)	< 1.8s on the same simulated connection
Time To Interactive (TTI)	< 3.5s
Cumulative Layout Shift (CLS)	< 0.1 — all images/media declare explicit dimensions (Part VII)
Largest Single Image Size	< 250KB (Hero backgrounds); < 200KB all other hero-tier images
Maximum JavaScript Size (per page, gzipped)	< 150KB total across global.js + page-specific script
Maximum CSS Size (per page, gzipped)	< 80KB total across global.css + page-specific stylesheet
Lazy Loading Strategy	Native <code>loading="lazy"</code> on all below-the-fold imagery; Skeleton Loader (Part III) shown during fetch to preserve layout stability
Image Optimisation Strategy	WebP primary format with compression targets per Part VII; responsive <code>srcset</code> per breakpoint
Font Optimisation Strategy	Maximum 2 font families / 4 total weights; <code>font-display: swap</code> with an immediate system-font fallback to avoid invisible-text flashes; fonts subset to required character sets where feasible
Expected Lighthouse Scores	Performance ≥ 90, Accessibility ≥ 95, Best Practices ≥ 95, SEO ≥ 95 (mobile, simulated throttling)

Part XI — Functional Behaviour Specification (Global Rules)

Per-element, per-page functional behaviour is documented in Part IV. This chapter defines the global rules that apply across every button, link, form, card, navigation item, filter, and search bar site-wide, so no interactive element anywhere on the site behaves ambiguously.

Rule	Specification
Outbound Link Behaviour	Every link leaving the site domain (Live Site, Source Code, social profiles, verification URLs) opens in a new tab with <code>rel="noopener noreferrer"</code> ; internal links never open a new tab.
Broken Reference Handling	Any internal link whose target does not exist resolves to the 404 page; this must never occur at launch (FR-12) and is a release-blocking defect if found.
Loading State Rule	Any control that triggers an asynchronous action (form submit) must show a Loading state (Part III Button spec) and must be disabled for the duration to prevent duplicate submission.
Empty State Rule	Any filterable/searchable collection (Projects, Blog) must render a dedicated Empty State with a recovery action when zero results match; it must never render a blank grid silently.
Failure State Rule	Any control whose action can fail (form submit, clipboard copy) must show a distinct, non-blaming failure message and must never fail silently.
Optional-Content Omission Rule	Where a content field is genuinely optional and absent for a given entry (e.g. a project with no live URL), the corresponding UI element is omitted entirely rather than rendered disabled or broken — see Part IV Section 21.5 for the canonical example.
Debounce Rule	Any live-filtering text input (Search Bar) debounces at 150ms before applying a filter, to avoid layout thrash on fast typing.
Single-Open Rule	Any accordion-pattern component (FAQ) allows only one open item at a time.
Animation Fire-Once Rule	Any scroll-triggered reveal animation fires exactly once per page view; it never re-triggers on re-scroll.
New-Tab Confirmation Rule	Controls that leave the site (outbound links) are visually distinguished (icon or hover-state cue) so the visitor understands they are leaving before clicking.

Part XII — Acceptance Criteria (Global Summary)

Per-page acceptance criteria are defined throughout Part IV. This chapter summarizes the site-wide, cross-cutting criteria that gate release readiness independent of any single page.

Category	Criteria (all must pass before launch)
Navigation	Every one of the 9 primary routes and all error routes are reachable from the global Navbar/Footer; no dead links anywhere on the site (FR-12).
Responsive Layout	Every page renders correctly with no horizontal scroll, no overlapping elements, and correct column counts at all four breakpoints, tested at the exact boundary widths (768px, 1024px, 1440px) per v1.0 recommendation.
SEO	Every indexable page has a unique meta title/description, correct canonical URL, valid Open Graph/Twitter Card data; <code>sitemap.xml</code> lists every indexable page and excludes all error pages; <code>robots.txt</code> correctly disallows <code>/html/errors/</code> .
Accessibility	WCAG 2.1 AA contrast and keyboard-navigability verified across all interactive elements per Part VIII; automated accessibility audit (e.g. axe) returns zero critical/serious issues.
Images Optimised	Every image meets its Part VII format/compression/dimension targets; all below-the-fold images are lazy-loaded; CLS budget (<0.1) is met.
Animations Working	Every animation in the Part II/VI motion inventory fires with the specified trigger/duration/easing; all animations correctly suppress under reduced-motion.
Forms Validated	Contact form validates every field per Part III Form spec; success and failure states both function correctly; no duplicate-submission possible.
Components Reusable	No page defines a one-off component style outside the Part III inventory and Part II design tokens (Non-Functional Requirement: Visual Consistency).
No Broken Links	Automated crawl of the built site returns zero 404s among internal links.
Performance Targets Achieved	All Part X metrics met on a simulated mid-range mobile device for every primary page.

Part XIII — Future Scalability (Expanded)

The growth-vector table from v1.0 Section 24 (Unlimited Projects/Blogs/Certificates, Future Themes, CMS/Backend/Auth/Admin/Database/API Integration, Search, Filtering, Pagination, Internationalisation) remains fully binding. This chapter adds implementation-level guidance for the vectors that most directly affect near-term maintainability.

Growth Vector	Implementation Guidance
Unlimited Projects / Blogs / Certificates	Governed entirely by the template-duplication model (v1.0 Section 15/16) and the Card family components (Part III Section 20.3); no page-level redesign is ever required to add an entry — only a new template-derived file and one new listing/grid entry.
Pagination (future)	Once Projects or Blog exceeds roughly 24–30 cards, introduce pagination or "Load More" using the existing Card grid and Filter/Search components unchanged; the grid component itself is already pagination-ready by virtue of rendering from an array.
Search Functionality (future, site-wide)	The existing per-page Search Bar (Part III) can be generalized to a site-wide search overlay without new visual components — only a broader index (projects + blogs + static pages) behind the same input pattern.
Future Themes (e.g. light mode)	Because every color/spacing/type value is a Design Token (Part II Section 16), a new theme is a token-set swap, not a rewrite; component markup and motion timing remain unchanged.
CMS / Backend Integration	The strict content/presentation separation (templates vs. per-page CSS/JS) means a future CMS maps directly onto existing template fields (Appendix D Content Inventory Checklist becomes the CMS schema starting point).
Authentication / Admin Dashboard	The 401/403 error pages exist today for architectural completeness; a future auth layer can gate an admin dashboard without touching any public-facing page or component.
Internationalisation (future)	All copy currently lives in static HTML per template; a future i18n pass would externalize copy into per-locale content files consumed by the same templates — no component or layout change required.

Part XIV — Appendices

Appendix A — Glossary

Term	Definition
PRD	Product Requirements Document — this document; the authoritative specification for what must be built.
Information Architecture (IA)	The structural organization of content and how it relates and connects across the site.
Design System	The single shared set of colors, typography, spacing, motion, and component rules every page draws from (Part II, Part III).
Design Token	A named, reusable value (color, spacing, etc.) referenced by code instead of a hard-coded literal.
Breakpoint	A defined screen-width range at which the layout adapts (v1.0 Section 20).
Scroll Reveal	An animation pattern where content fades and rises into view the first time it enters the viewport.
Glassmorphism	A translucent, blurred visual style used sparingly — primarily the mobile navigation overlay.
Template Page	A blank, structurally complete HTML file duplicated to create new content entries.
WCAG	Web Content Accessibility Guidelines — the standard this site targets at Level AA.
LCP / FCP / TTI / CLS	Largest Contentful Paint / First Contentful Paint / Time To Interactive / Cumulative Layout Shift — performance metrics governed in Part X.
Structured Data	Machine-readable metadata (e.g. schema.org markup) that helps search engines understand page content.
Debounce	Delaying an action (e.g. live filtering) until input has paused briefly, to avoid excessive re-computation.
Skeleton Loader	A placeholder matching the eventual content's shape, shown while that content loads.
ARIA	Accessible Rich Internet Applications — attributes that expose custom UI semantics to assistive technology.

Appendix B — Assumptions

- The stakeholder supplies final copy (bio text, project descriptions, blog articles) — this PRD defines structure and behaviour, not final wording, though Part VI governs voice/tone.
- The stakeholder supplies a resume file, favicon source, and any personal photography/portraits used in About and Home.
- No user accounts, authentication, or backend persistence exist at this stage; 401/403 error pages exist for architectural completeness and future access-controlled areas, not a current login system.
- The contact form's submission handling mechanism (e.g. a third-party form endpoint) is an implementation detail left to the development team, provided the visitor-facing behaviour in Part IV Section 21.9 is preserved.

- All performance targets in Part X assume static hosting with a CDN in front of the site; targets may need re-baselining if hosting changes materially.

Appendix C — Open Questions for Stakeholder Sign-off

#	Question
1	What is the final list and priority order of projects to feature on Home (Part IV Section 21.1)?
2	What is the initial category taxonomy for Projects and Blog (Part IV Sections 21.4, 21.6)?
3	What specific pricing tiers or ranges should appear on the Hire Me page (Part IV Section 21.8)?
4	Which social/contact channels take priority if screen space is constrained on mobile (Part IV Section 21.9)?
5	Should Projects/Blog filter and search state persist via URL query parameters (a documented future enhancement, Part IV Section 21.4) or remain session-local at launch?
6	Exact hex/rgba values for each Design Token (Part II Section 16) — this document defines character and contrast requirements; final values require stakeholder visual sign-off.

Appendix D — Content Inventory Checklist (To Be Maintained by Stakeholder)

Content Type	Required Fields per Entry
Project	Title, category, tags/technologies, thumbnail, gallery images, tagline, overview, problem statement, features, challenges, solutions, live URL, source URL, future improvements
Blog Post	Title, category, tags, publish date, cover image, excerpt, reading time, body content, author (default: owner)
Certificate	Title, issuing organization, date issued, credential image/link
Achievement	Title, context sentence, date

Appendix E — Future Roadmap

Horizon	Potential Enhancement
Near-term	Populate initial project and blog content per Appendix D; finalize pricing placeholders on Hire Me.
Mid-term	Introduce lightweight analytics to inform Home Featured selections; explore a light-mode theme variant via token swap (Part XIII).
Long-term	Evaluate a headless CMS or lightweight backend once content volume makes static template duplication cumbersome (Part XIII); evaluate localization/multi-language support.

Appendix F — References

- WCAG 2.1 (W3C Web Content Accessibility Guidelines), Level AA conformance target.
- Web Vitals (LCP, FCP, CLS, TTI) definitions as used throughout Part X.

- schema.org vocabulary — Person, Article, and CreativeWork types referenced in Part IX.
- WAI-ARIA Authoring Practices — Tabs, Accordion, and Dialog (Modal) interaction patterns referenced in Part III.

CLOSING NOTE: This document (v2.0) supersedes PRD v1.0 in full and is intended to remain the single, implementation-ready source of truth for the Ammlan Rath Portfolio Website. Any future change in scope, design direction, or architecture should be reflected back into this PRD before implementation begins.